Port City International University

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Report on

Sending messages using Bluetooth Module.

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Course code : CSE 334

Course Title : Computer Peripherals and Interfacing Sessional

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Report title: Sending messages using Bluetooth Module.

Objective: The objective of this experiment is to understand the working principle of wireless communication between an Arduino UNO and a mobile device using an HC-05 Bluetooth module. Through this experiment, learners will gain practical experience in sending and receiving data wirelessly, monitoring the received data on the Arduino Serial Monitor, and establishing two-way communication between a mobile device and the Arduino. This helps students understand serial communication, Bluetooth interfacing, and basic microcontroller-based data exchange systems.

Required Instruments:

1. Arduino Uno R3	1
2. HC-05 Bluetooth module	1
3. Breadboard	1
4. Jumper wires	as required
5. Serial Bluetooth Terminal app installed	1

Procedure:

1. Connect the HC-05 Bluetooth module to the Arduino UNO:
    - VCC → 5V
    - GND → GND
    - TX → Arduino pin 10
    - RX → Arduino pin 11
2. Placed all components properly on the breadboard.
3. Uploaded the Bluetooth communication code to the Arduino board.
4. Open the Serial Monitor to observe incoming data from the mobile device.
5. On mobile device, open the Serial Bluetooth Terminal app and connect to the HC-05 module.
6. Send data from the mobile app and observe it appearing on the Arduino Serial Monitor.

Source Code:

```
#include <SoftwareSerial.h>

SoftwareSerial BT(10, 11);

void setup() {

Serial.begin(9600);
```

```
BT.begin(9600);
Serial.println("Bluetooth Module Ready. Waiting for commands...");
}
void loop() {
if (BT.available()) {
char c = BT.read();
Serial.print("Received: ");
Serial.println(c);
}
if (Serial.available()) {
BT.write(Serial.read());
}
}
```

Output:

Software:

Figure1: Sending data in Serial Bluetooth Terminal.

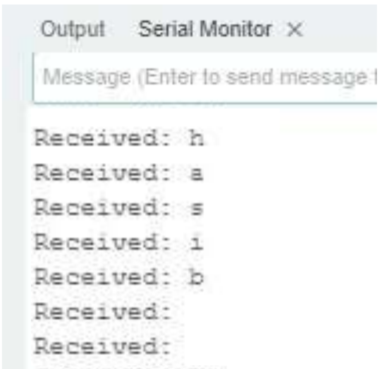

Figure2: receiving data in Serial Monitor.

Hardware:

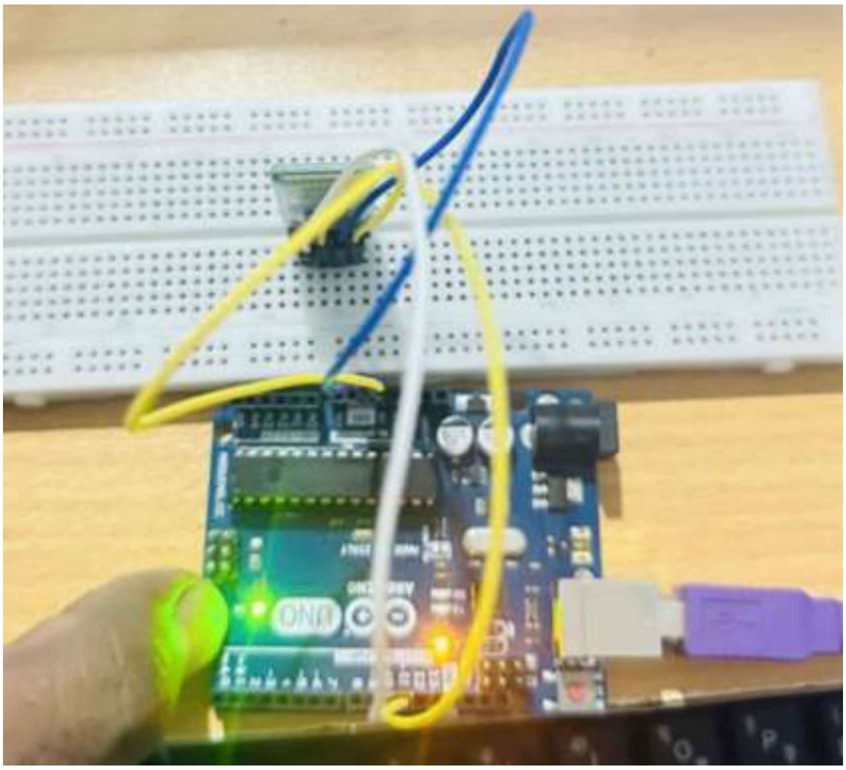

Figure 2: Set up of Bluetooth Module.

Discussion: In this experiment, I used an HC-05 Bluetooth module to send and receive messages between an Arduino UNO and a mobile device. The Arduino was able to receive data sent from the mobile app and display it on the Serial Monitor, and it could also send data back to the mobile device. This showed me how software serial allows the Arduino to communicate with the module without interfering with the USB connection. This experiment helped me understand basic Bluetooth interfacing and practical serial communication in microcontroller projects.